Tuberculosis Detection Beyond Dataset Bias: Exploring Transfer Learning and Synthetic Generation

Mannan Naidu
Department of Electronics and Computer Science Engineering
Vidyalankar Institute of Technology
Mumbai, India
mannan.naidu@vit.edu.in

Sameer Shelar
Department of Electronics and Computer Science Engineering
Vidyalankar Institute of Technology
Mumbai, India
sameer.shelar@vit.edu.in

Anoushka Joshi
Department of Biomedical Engineering
Vidyalankar Institute of Technology
Mumbai, India
anoushka.joshi@vit.edu.in

Dr. Uma Jaishankar
Department of Electronics and Computer Science Engineering
Vidyalankar Institute of Technology
Mumbai, India
uma.jaishankar@vit.edu.in

Abstract

The purpose of this study is to provide an exhaustive examination of various representation learning techniques as they pertain to the identification of tuberculosis (TB) in chest X-rays. The methodology will explore two different approaches; a numerical feature-based method using transfer learning along with latent representations derived from an autoencoder, synthetic augmentation of the feature space, and an image generation methodology utilizing specialized region-of-interest (ROI) knowledge via diffusion-based synthetic sample creation. To determine the relative performance of each methodology, the TBX11K Simplified dataset will be used for model development, and independent validation datasets will be utilized to evaluate generalization performance. Cross-dataset evaluation on an independent 4,200-image tuberculosis dataset demonstrates that the diffusion-enhanced ROI model outperforms all comparative architectures, achieving 98.45% accuracy, 94.46% recall, and 95.32% F1-score. In comparison, Dual-Stream Fusion achieved 96.10% accuracy (90.54% recall, 88.56% F1-score), Context-Aware Diffusion reached 94.97% accuracy with moderate recall, and the ROI-only model attained the highest recall (96.79%) with a precision trade-off. The baseline CNN achieved 93.51% accuracy and its lower recall limits screening suitability. These findings confirm the superiority and robustness of diffusion-guided ROI feature regularization under dataset shift conditions.
Keywords: Medical imaging diagnosis, data imbalance solutions, tuberculosis screening, diffusion models, synthetic augmentation, diagnostic imaging AI, representation learning methods.

1. Introduction

The World Health Organization estimates that millions of people will develop and die from tuberculosis each year, even though there are effective treatments available to those who are diagnosed early [1]. One of the most commonly used ways to diagnose tuberculosis is by taking a chest x-ray. However, the interpretation of chest X-ray images requires significant expertise and is often associated with inter-observer variability, especially in resource-constrained settings [2]. Significant improvements have been reported recently using deep learning approaches to improve the automatic analysis of medical images, especially for the detection of thoracic diseases [3]. CNNs have shown promising results in the classification of chest X-ray images [4], while Vision Transformers have recently gained popularity as potential alternatives to CNNs [7], especially for their ability to model long-range contextual dependencies [15].

Despite the encouraging results of medical imaging models on controlled benchmarks, there is increasing evidence that many of the top-performing models have poor performance under distribution shift and cross-dataset evaluation [9] [18]. This has been attributed to the problem of shortcut learning [10].

For the problem of data imbalance and robustness, there have been attempts to leverage generative augmentation

methods. GAN-based methods have been used to generate medical images [12] [13], but recently, diffusion probabilistic models have been shown to have improved stability, diversity, and coverage on medical imaging tasks [11]. There is also evidence that the generated images may have an effect on the learning of the representation and the decision boundaries; however, their incorporation into ROI-aware transfer learning methods for TB detection is still an under-explored area. This study investigates two complementary approaches which are-

A. Numerical Feature Vector Methodology

We examine image-level TB classification without incorporating region-based information or synthetic data. This includes a standard baseline trained directly on full chest radiographs, as well as an enhanced variant that leverages learned feature representations transferred across models. These experiments aim to establish strong reference points and assess the extent to which discriminative numerical feature vectors alone can capture TB-related patterns under class imbalance.

B. Region of Interest based transfer learning

In the second direction, we train feature extractors on ROIs and then transfer this learning to a classifier head.

Synthetic generation of ROIs using diffusion is also explored to mitigate class imbalance.

2. Literature review

2.1 CNNs and Vision Transformers in Medical Imaging

Although CNNs remain the dominant approach in the analysis of chest radiographs, transformer models have the capacity to incorporate global contextual information. Recent reviews have shown that Vision Transformers and CNN-Transformer models have promising performance in various medical image analysis tasks, such as thoracic pathology classification [7] [8] [15]. However, transformer models have to be adapted well to the data to achieve good generalization.

2.2 Feature Extraction and Representation Learning

Transfer learning using CNNs pretrained on large-scale natural image datasets has been widely adopted in medical imaging due to limited labelled data availability. Residual networks introduced by He et al. [4] and other pretrained architectures provide strong initialisation and enable effective feature extraction for chest X-ray analysis [6].
Unsupervised feature learning methods, such as autoencoders, have also been investigated for chest X-ray analysis, primarily for anomaly detection and representation learning. While these approaches can model general anatomical structures, they often lack the discriminative power required for precise TB classification, especially when pathological patterns are subtle and overlap with other pulmonary abnormalities [10].

2.3 Region-of-Interest-Based (ROI) Learning

Tuberculosis-related abnormalities are often localised to specific lung regions, motivating ROI-based learning strategies. Prior work has shown that focusing on localised regions can improve feature relevance and interpretability by reducing background noise and emphasising pathological cues [14].

2.4 Class Imbalance and Data Augmentation

Class imbalance remains a major problem in TB detection. Conventional oversampling techniques such as SMOTE [5] and regularization-based augmentation strategies such as Mixup [16] have been explored to mitigate class imbalance and improve generalization in deep learning systems but are not effective in the high-dimensional feature space of medical images. GAN-based synthetic data augmentation has also been investigated, though its applicability is marred by instability issues and lack of diversity in the augmented data [13].

Several surveys have demonstrated the superior stability, diversity, and robustness of diffusion probabilistic models compared to GAN-based solutions [11]. Moreover, it has been shown that the use of diffusion model-generated synthetic data can help to achieve fairness and robustness in medical classification systems, avoiding the use of spurious information specific to the datasets used [12]. However, the application of diffusion model-based data augmentation in the ROI-guided TB detection problem has not been adequately investigated.

2.5 Research Gap

(i) the existing methods have not sufficiently evaluated the CNN-based detection of TB under practical conditions of class imbalance;
(ii) the transfer-based methods have not sufficiently incorporated the ROI-based representation learning;
and (iii) the diffusion-based augmentation of the training dataset has not been sufficiently explored to address the issue of distribution shift and representation bias in the classification of chest X-ray images [9] [10].

3. Data and variables

3.1 Data and Characteristics

Our experimental basis is the TBX11K Simplified dataset [17]. The TBX11K dataset was created from a total of 8,811 chest radiographs with annotations done by a board-certified radiologist. The dataset is considerably imbalanced due to a class distribution (number of normal radiographs, TB positive) of 7600/1211 which results in a clinically-representative mixture of 6.28:1 normal – vs – TB examinations. All radiographs in the dataset are produced at a standard size (512 × 512 pixels), and all chest radiographs remain as single channels throughout the dataset.

3.2 Cross-Dataset Validation

Evaluation of generalisation will be done using the Tuberculosis Chest X-ray Dataset [18], which contains 4,200 images, of which 3,500 are normal and 700 are TB positive, as an independent validation to provide validation of robustness across multiple clinical data collections from various sources.

4. Methodology and model specifications

4.1 Baseline Model

Baseline model development used fully acquired chest radiographic images as described above without any type of regional support methods or artificial augmentations. ResNet-18 and ViT-B/16 architectures utilized pre-trained ImageNet-based parameters and were converted to grayscale images. Each model was trained for ten epochs using the Adam optimization technique with a learning rate set at 1×10^{-4}. All images undergo resizing to 224×224 pixel dimensions, with cross-entropy loss implementation excluding clasrebalancing considerations.

4.2 Numerical Feature Vector Methodology

4.2.1. Feature Extraction Framework

A convolutional autoencoder with 27M parameters learns 256-dimensional latent representations through reconstruction optimization. Despite achieving excellent reconstruction fidelity (MSE: 0.0017), subsequent analysis reveals limited diagnostic discriminative capacity.

4.2.2 ResNet50 Transfer Learning

Pre-trained ResNet50 extracts 2,048-dimensional feature vectors with all convolutional layers frozen, providing robust anatomical representations without overfitting to limited medical data.

4.2.3 Dual-Stream Feature Fusion Architecture

Our core contribution integrates complementary feature types:

$$F_{fused} = \text{concatenate}(f_{ResNet}, f_{AE}) \in \mathbb{R}^{2304}$$

Where, $f_{ResNet} \in \mathbb{R}^{2048}$ represents transfer-learned anatomical knowledge and $f_{AE} \in \mathbb{R}^{256}$ captures dataset-specific latent patterns. This synergistic combination leverages ResNet50's general visual understanding with the autoencoder's reconstruction-optimised representations, creating a comprehensive feature space that addresses limitations of individual modalities.

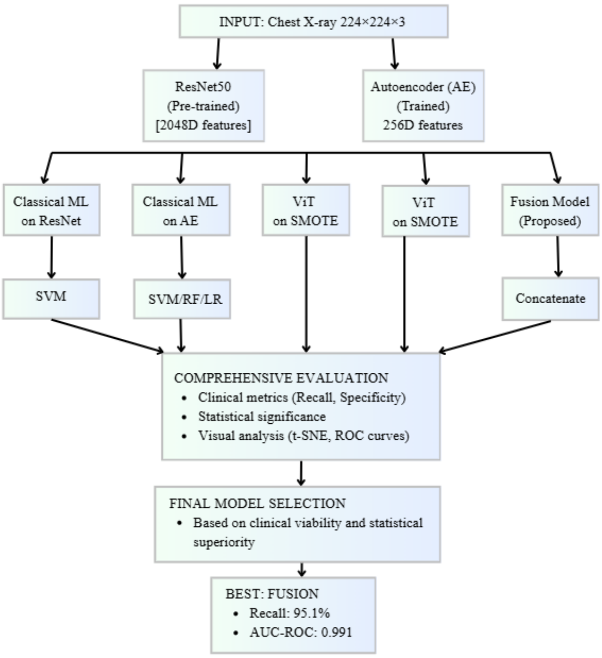

Fig. 4.2.3 Dual-Stream Feature Fusion Architecture

4.2.4 Class Imbalance Mitigation

Feature-level Synthetic Minority Over-sampling Technique (SMOTE) generates synthetic tuberculosis samples within the **2,304-dimensional fused feature space** through k-nearest neighbor interpolation (k=5). This preserves relational structures between feature types unlike pixel-level augmentation.

So the fusion model comprises of Autoencoder and ResNet50's extracted features, furthermore the features are concatenated and SMOTE is applied on it, to form the dual-stream fusion model.

4.2.5 Classification Models

- Baseline Models: Logistic Regression, Random Forest, Support Vector Machine
- Vision Transformer: Four transformer layers processing autoencoder features
- Fusion Classifier: Dense network (256→128→64→1) with dropout regularization trained with class weighting (Normal:1.0, TB:4.6)

4.3 Region of Interest based transfer learning

The training configuration employed an 80-20 train-validation split at the ROI level, patient-disjoint partitioning to prevent data leakage, and 10-epoch optimisation

4.3.1 ROI-Only Representation Learning (No Synthetic Images)

In the first variant, only real TB-positive regions of interest (ROIs) extracted from bounding box annotations are used for representation learning. Cropped ROIs are resized and normalized before being used to train an ROI feature extractor based on either a CNN or a Vision Transformer (ViT) backbone. Since ROIs correspond only to TB-affected regions, the extractor is trained as a disease-specific representation learner, rather than a discriminative classifier. The learned encoder weights are then transferred to the backbone of a global image-level classifier, while the classification head is initialized randomly. The resulting model is trained on full chest X-ray images using patient-disjoint splits, without access to ROI information during inference. This variant serves as a region-aware baseline, isolating the contribution of ROI-based representation learning without synthetic augmentation.

Fig. (i)

Fig. (ii)

Fig. (iii)

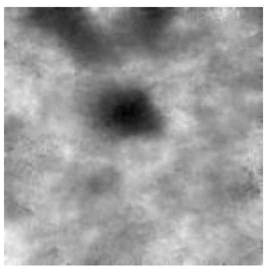

Fig. (iv)

Fig. 4.3.1 Synthetic Image Generation where, Fig. (i) Raw Chest X-Ray Image, Fig. (ii) Extracted ROI, Fig. (iii) Pre-processed ROI and Fig. (iv) Synthetic Lesion

4.3.2 Raw ROI–Based Synthetic Image Generation (Diffusion on Raw ROIs)

The second variant improves ROI-based representation learning by using synthetic TB ROIs created through a diffusion model. A denoising diffusion probabilistic model (DDPM) is trained directly on raw, unprocessed ROIs to learn the patterns associated with TB-positive regions.

After training, the diffusion model generates synthetic ROIs. These are mixed with real ROIs to create an augmented training set. This combined dataset, with a 3:1 ratio of real to synthetic, is used to train the ROI feature extractor. Like the ROI-only variant, the extractor's learned weights transfer to a global image-level classifier, which trains only on full images.

This method tests whether using diffusion-based augmentation can enhance representation learning when it focuses on raw ROI distributions.

4.3.3 Preprocessed ROI–Based Synthetic Image Generation (Diffusion on Normalized ROIs)

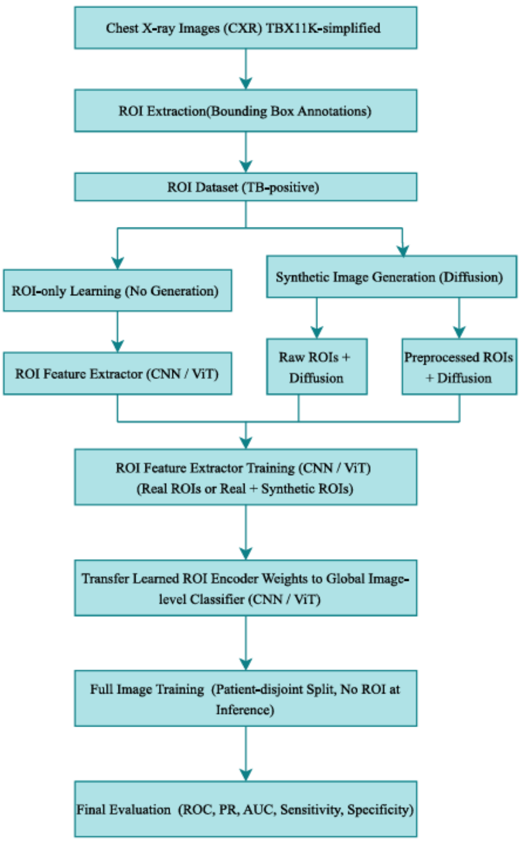

Fig. 4.3.3 Region of Interest based transfer learning Architecture

The third variant improves diffusion-based augmentation by adding ROI preprocessing before generating synthetic images. Before diffusion training, TB ROIs go through contrast normalization to lessen acquisition-specific differences and highlight disease-related structures.

The diffusion model is trained on these preprocessed ROIs. Then, it generates synthetic samples in the normalized ROI space. Synthetic and real preprocessed ROIs are combined to train the ROI feature extractor. As with earlier variants, the trained encoder is used in the global image-level classifier, which is trained on complete images with patient-disjoint splits.

This variant tests the idea that normalizing distribution before diffusion training enhances the effectiveness of synthetic ROIs for later representation learning.

4.4 Cross-Dataset Testing Methodology

To evaluate model generalization beyond dataset-specific bias, cross-dataset testing was performed using the Tuberculosis Chest X-ray dataset.

4.4.1 Preprocessing: All images were processed using a unified pipeline. Images were resized to 224 × 224, converted to grayscale (1-channel), transformed into PyTorch tensors, and normalized using a mean and standard deviation of 0.5.

4.4.2 Dataset Splitting: The dataset was split into 20% finetuning and 80% independent testing subsets using stratified sampling to preserve class balance between *Normal* and *Tuberculosis* cases. A fixed random seed (42) ensured reproducibility across all experiments.

4.4.3 Model Architecture: All models were based on a ResNet-18 backbone. The first convolutional layer was modified to accept single-channel input, and the final fully connected layer was adapted to output two classes. Depending on availability, models were initialized with either pre-trained or randomly initialized weights.

4.4.4 Finetuning Setup: Model fine-tuning proceeded for five epochs employing Adam optimization with 0.001 learning rate specification and cross-entropy loss implementation. A batch dimension of 32 facilitated training progression. This standardised configuration enabled equitable comparison across methodologies while isolating generalisation performance under cross-dataset evaluation conditions.

5. Results

CNN outperformed ViT in every metric due to lack of a large enough dataset for ViT. A conservative ratio (1:1) was maintained during epochs to avoid overfitting on synthetic patterns so that synthetic data is used only for regularization. Cross testing was performed on the better performing CNN model for all approaches.

5.1 Cross Testing metrics and interpretation

The baseline CNN, which is trained only on full images, faces a significant drop in recall during cross-testing, even though it has high precision. This highlights the poor generalization of baseline model.

The ROI-only model, which is the base for transfer learning approach, has a very high recall. This helps us conclude that training a feature extractor on ROIs improves identification of TB lesion in comparison to the baseline model. But its low precision highlights the high number of false positives which could be caused by interfering noise and small training set.

The Context-aware diffusion model features high precision but lower recall. The model missed out on lot of positive cases and hence cannot be used for actual screening. But better precision over the previous model provides a base for further investigation in this direction.

The Dual Stream Fusion Model has a 95.1% recall/98.6% accuracy on the primary TBX11K dataset, thereby validating that integrating at the feature level is an effective substitute for augmenting through creating new pixels in an augmented image. In contrast, the Autoencoder + ViT method exhibited an extremely low recall of 8.2% despite having an accuracy of 82.1%, which highlights the gap between diagnostic discriminative ability and the quality of reconstruction that drives our fusion framework.

The Diffusion (Preprocessed ROIs) model shows the best overall performance across datasets,. It has the highest accuracy and F1-score. This suggests that using diffusion-based augmentation on normalized ROIs helps improve generalization across different datasets with different acquisition characteristics after a little fine tuning.

Table 5.1 Cross-Dataset Generalization Performance (CNN) and Potential real-world applications in clinical settings

Model	Accuracy	Precision	Recall	F1-score	Key Strength	Optimal Clinical Application
Baseline CNN	0.9351	0.9914	0.6161	0.759	Interpretability	Educational support, supervised settings
ROI Only	0.9405	0.7486	0.9679	0.844	Maximum sensitivity	High-stakes pre-screening (airports, immigration)
Context-aware Diffusion	0.9497	0.9827	0.7107	0.824	Precision	Confirmatory diagnosis, specialist referral
Dual-Stream Fusion	0.961	0.8667	0.9054	0.885	Balanced utility	Primary screening, rural clinics
Diffusion (Preprocessed ROIs)	**0.9845**	**0.9618**	**0.9446**	**0.953**	**Generalization**	**Multi-region programs, varied equipment**

Table 5.2 Test metrics (ViT)

Model	Accuracy	Precision	Recall	F1-score
Baseline ViT	0.9615	0.8582	0.872	0.865
ROI Only	0.9570	0.9833	0.708	0.823
Context-aware Diffusion	0.9350	0.7846	0.825	0.804
Autoencoder +SMOTE+ ViT	0.8209	0.1786	0.820	0.112
Diffusion (Preprocessed ROIs)	**0.9491**	**0.9108**	**0.760**	**0.828**

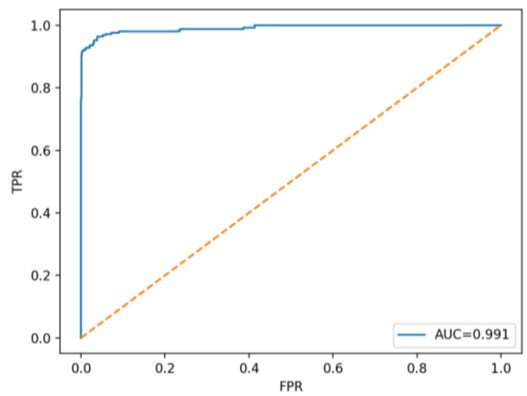

Fig. 5.1 AUC (Diffusion Preprocesses ROIs)

5.2 Interpretation for superiority of Diffusion (Preprocessed ROIs model)

Diffusion-generated ROIs enhance classification performance by providing a means to alter levels of representation rather than simulating the characteristics of lesions that are more realistically present. The diffusion-generated ROIs can be used as feature-space regularizers for developing models that depend on recognising predictable patterns in cases of class imbalance by modifying how the models determine patterns in these cases. The ROIs reduce feature-space texture and acquisition bias by eliminating anatomically based features as well as scanner-specific patterns of intensity. By deleting the physical anatomical features and scanner-specific intensity patterns, these examples reduce the potential for misleading correlations that could be misused by convolutional and transformer-based models when trained on chest X-rays. This improves learning stability and develops more consistent representations in models; therefore, cross-dataset evaluations of performance are improved. The addition of diffusion-generated ROIs reduces the classifier's decision boundary delineation.

Although these samples seem to have visually simple representations, they occur in areas of the learned feature space with low density, thus, they would be examples of well-structured hard examples. The effect of these hard examples is that the decision surface becomes tighter, there will be less overconfidence on borderline cases and there will be greater recall stability with an undersized area under the ROC curve. Another benefit of using diffusion-generated ROIs for implicit augmentation is that by using implicit augmentation, the diffusion-generated ROIs can add no additional label noise. Unlike randomly generated transformations (at the image level), the diffusion-generated ROIs remain consistent across all positive

class examples while increasing the amount of effective support that exists for the positive class in latent space.
Additionally, diffusion-generated ROIs serve as reference points to control representation learning which helps stabilize representation learning as opposed to providing more clinically significant instances of diseases. In conclusion, the improvement in the performance is mainly due to the improvement in stability of representation and regularization of decision boundary, not due to the addition of clinically realistic disease patterns.

5.3 Feature-Based Representation Insights

Our numerical feature investigations reveal parallel regularization mechanisms:

- Feature-Space SMOTE operates similarly to diffusion-generated ROIs, creating synthetic samples in latent space that act as hard examples to sharpen decision boundaries.
- The reconstruction-diagnosis disparity (Auto-encoder: 0.0017 MSE vs 0.082 recall) demonstrates that visual realism (reconstruction quality) is not diagnostic utility, mirroring the diffusion finding that synthetic realism is not classification improvement.

5.4 Limitations

The images generated by diffusion do not correspond to clinically meaningful TB lesions and thus do not add pathological diversity. They often lack anatomical structures and the improvements are due to regularization. Training on global images compromises on the tb lesions and ROI based training results in lack of anatomical structure. Diffusion models are computationally expensive and consume enormous amounts of time for synthetic data generation. Diffusion models are also data hungry and there is an urgent need to have an open source comprehensive database on tb lesions to make such studies more accessible.

6. Conclusion

This investigation evaluates various approaches for tuberculosis detection trying to mitigate class imbalance. CNN outperforms ViT due to the small dataset size. Thus, cross-testing for CNN was performed and various approaches were compared. The baseline model, which had overfitted on “tbx11k-simplified” dataset had poor performance on cross-testing, concluding its poor generalization. The ROI-based transfer learning approach significant improvement in recall value showing signs for higher tb detection. When synthetic ROIs are generated using diffusion, performance drops due to poor quality synthetic images. The diffusion-based (preprocessed) transfer learning approach was the best model performing well across metrics. It has excellent cross-dataset testing generalization and can be used for reliable tb screening. The synthetic generation and the parallel numeric vector approach confirms that class imbalance in dataset can be mitigated by representational regularizers instead of accurate clinical sample.

7. References

[1] World Health Organization, *Global Tuberculosis Report 2023*, Geneva, Switzerland, 2023.
[2] J. A. Esmail et al., “Tuberculosis diagnostics: current challenges and future directions,” *Clinical Infectious Diseases*, vol. 69, no. 1, pp. 152–159, 2019.
[3] A. Lakhani and B. Sundaram, “Deep learning at chest radiography: automated classification of pulmonary tuberculosis,” *Radiology*, vol. 284, no. 2, pp. 574–582, 2017.
[4] Kaiming He et al., “Deep residual learning for image recognition,” in *Proceedings of CVPR*, 2016.
[5] N. V. Chawla et al., “SMOTE: Synthetic minority over-sampling technique,” *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
[6] J. Yosinski et al., “How transferable are features in deep neural networks?” in *Advances in Neural Information Processing Systems*, 2014.
[7] A. Khan et al., “Transformers in medical image analysis: A review,” Medical Image Analysis, vol. 79, 2022.
[8] E. U. Henry, O. Emebo, and C. A. Omonhinmin, “Vision Transformers in Medical Imaging: A Review,” *arXiv preprint*, arXiv:2211.10043, 2022.
[9] Q. Wang et al., “Distribution shift in medical image analysis: A survey,” Medical Image Analysis, vol. 85, 2023.
[10] R. Geirhos et al., “Shortcut learning in deep neural networks,” Nature Machine Intelligence, vol. 2, pp. 665–673, 2020.
[11] A. Croitoru et al., “Diffusion models in medical imaging: A comprehensive survey,” Medical Image Analysis, vol. 88, 2023.
[12] I. Ktena, O. Wiles, I. Albuquerque, S.-A. Rebuffi, R. Tanno, A. G. Roy, S. Azizi, D. Belgrave, P. Kohli, T. Cemgil, A. Karthikesalingam, and S. Gowal, “Generative models improve fairness of medical classifiers under distribution shifts,” Nature Medicine, vol. 30, pp. 1166–1173, Apr. 2024.
[13] I. Goodfellow et al., “Generative adversarial nets,” in *Advances in Neural Information Processing Systems*, 2014.
[14] N. Rashid, M. A. F. Hossain, M. Ali, M. I. Sukanya, T. Mahmud, and S. A. Fattah, “AutoCovNet: Unsupervised feature learning using autoencoder and feature merging for detection of COVID-19 from chest X-ray images,” *Biocybernetics and Biomedical Engineering*, vol. 41, no. 4, pp. 1685–1701, Oct.–Dec. 2021.
[15] A. Dosovitskiy et al., “An image is worth 16×16 words: Transformers for image recognition at scale,” in *International Conference on Learning Representations*, 2021.
[16] H. Zhang et al., “Mixup: Beyond empirical risk minimization,” in *ICLR*, 2018.
[17] L. Liu, H. Zhao, Y. Li, and X. Wang, “TBX11K: A large-scale benchmark dataset for tuberculosis detection and localization in chest X-rays,” *IEEE*

Transactions on Medical Imaging, vol. 42, no. 7, pp. 1985–1996, 2023.
[18] T. Rahman, M. E. H. Chowdhury, A. Khandakar, K. R. Islam, K. F. Islam, Z. B. Mahbub, and M. A. Kadir, “Tuberculosis (TB) Chest X-ray Dataset,” *Kaggle*, 2020.